
REQUIREMENTS NOT MET

N/A

VIDEO FILE LINK

https://youtu.be/lf14VnVAcK0?si=UIVwC0SdvP7_AuV1

PROBLEMS ENCOUNTERED

I had Soor help me through Slack and realized that I was doing POS wrong, I was assuming that we were saying a variable was true when it was 1, but through his help found that is only true for SOP, for POS we look for where the variables are 0 to be true.

FUTURE WORK/APPLICATIONS

This is the most practical so far, 7-segment LEDs are used all around the world, for appliances, watches, anything that can count, etc. You could use these 7-segment decoder to convert from binary to hexadecimal, drive a display, which are fundamental skills in electronics.

PRE-LAB QUESTIONS OR EXERCISES

SECTION 1

Q1:

Table 1: Truth Table for 2-input MUX

S0	D0	D1	Y
0	0	-	0
0	1	-	1
1	-	0	0
1	-	1	1

Row 1: $(\neg S0 * D0) = 1$

Row 3: $(S0 * D1) = 1$

SOP: $Y = (\neg S0 * D0) + (S0 * D1)$

Row 2: $(\neg S0 + \neg D0) = 0$

Row 4: $(S0 + \neg D1) = 0$

POS: $Y = (\neg S0 + \neg D0) * (S0 + \neg D1)$

Table 2: K-Map for 2-input MUX.

$(S0)/(\neg D0 * D1)$	00	01	11	10
0	0	0	1	1
1	0	1	1	0

Groupings

For Red: $\neg S0 + \neg D0 \leftarrow \text{Zeroes}$

For Orange: $\neg S0 * D0 \leftarrow \text{Ones}$

For Green: $S0 + \neg D1 \leftarrow \text{Zeroes}$

For Blue: $S0 * D1 \leftarrow \text{Ones}$

SOP: $Y = (\neg S0 * D0) + (S0 * D1) \leftarrow \text{Ones}$

POS: $Y = (\neg S0 + \neg D0) * (S0 + \neg D1) \leftarrow \text{Zeroes}$

ALREADY SIMPLIFIED!

Table 3: Truth Table for 2-input MUX.

D0	D1	S0	/S0	(/S0*D0)	S0*D1	Y
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	1	0	1	0	0	0
0	1	1	0	0	1	1
1	0	0	1	1	0	1
1	0	1	0	0	0	0
1	1	0	1	1	0	1
1	1	1	0	0	1	1

Table 4: Voltage Table for 2-input MUX.

D0(L)	D1(H)	S0(H)	Y(L)
L	L	L	L
L	L	H	H
L	H	L	L
L	H	H	L
H	L	L	H
H	L	H	H
H	H	L	H
H	H	H	L

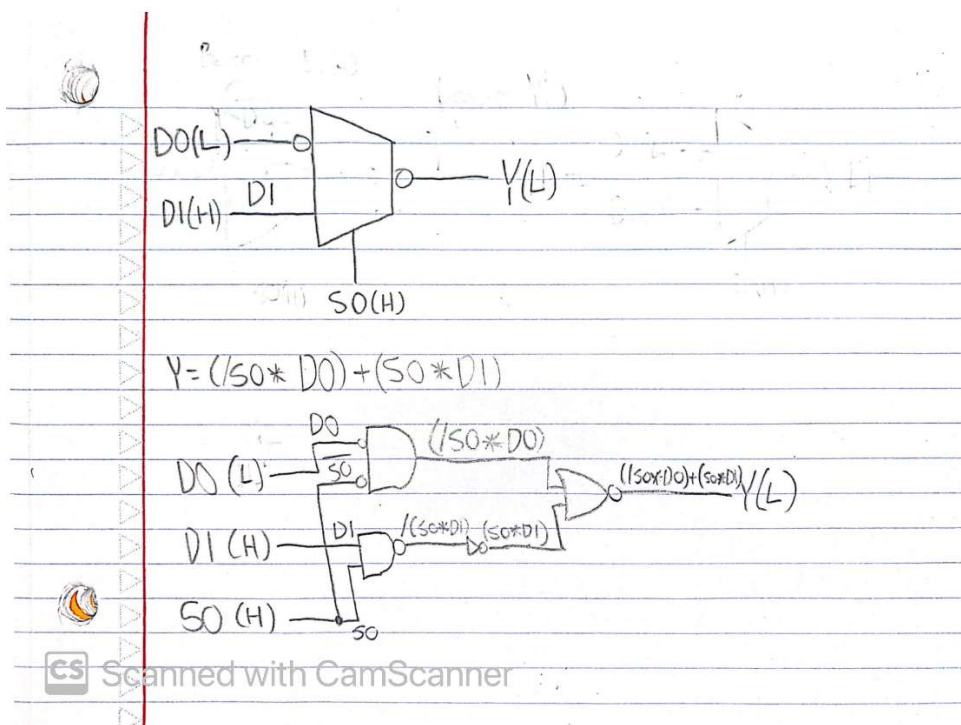


Figure 1: MUX Functional Block Diagram & Circuit Block Diagram

Lab 2, Part 1

Name: Evan Baessler

Class #: 11081

PI Name: Tyler Ruble

Description: $Y = (/S0 * D0) + (S0 * D1)$

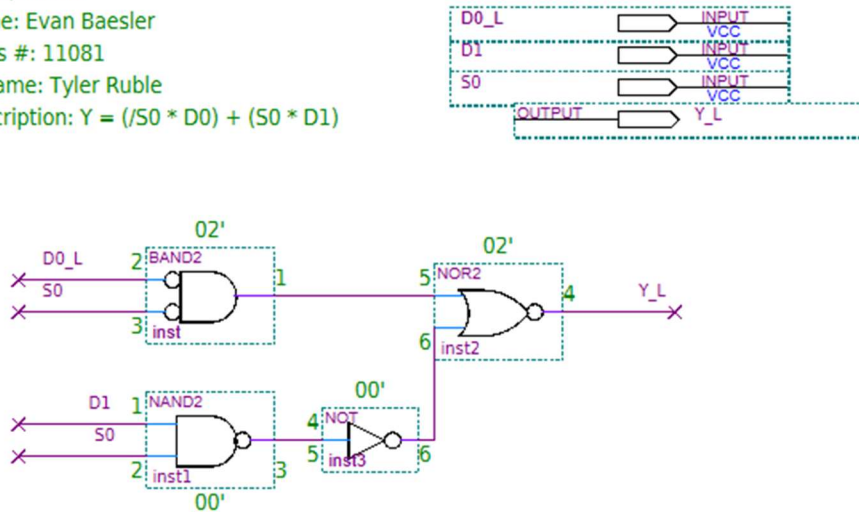
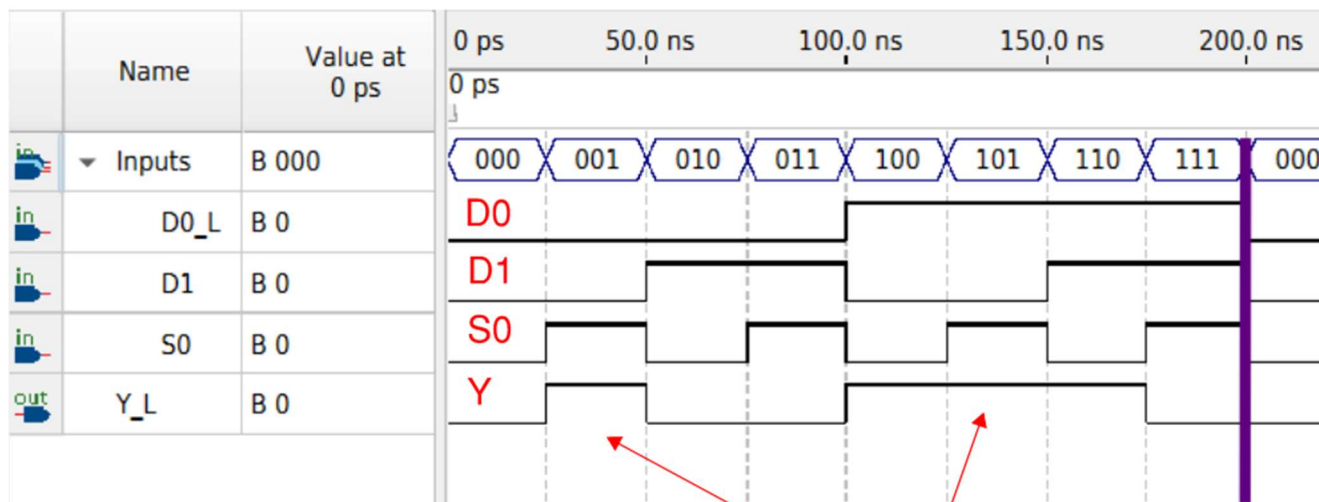


Figure 2: Quartus Wiring Diagram for $Y = (/S0 * D0) + (S0 * D1)$



We have an active-high at 001, 100-110, as shown on our voltage table. These voltages are matching based on whether the signal is active high or active low, therefore they should match.

Figure 3: Annotated Waveform for $Y = (/S0 * D0) + (S0 * D1)$

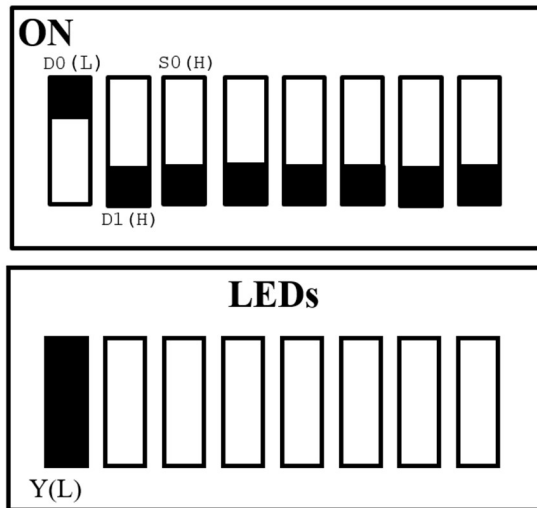


Figure 4: Switch & LED Diagram (at HHH.)

SECTION 2

Table 5: Truth Table for 4-input MUX with wildcards.

S0	S1	D0	D1	D2	D3	Y
0	0	0	-	-	-	0
0	0	1	-	-	-	1
0	1	-	0	-	-	0
0	1	-	1	-	-	1
1	0	-	-	0	-	0
1	0	-	-	1	-	1
1	1	-	-	-	0	0
1	1	-	-	-	1	1

Row 2: $\neg S0 * \neg S1 * D0 = 1$

Row 4: $\neg S0 * S1 * D1 = 1$

Row 6: $S0 * \neg S1 * D2 = 1$

Row 8: $S0 * S1 * D3 = 1$

SOP $\rightarrow Y = (\neg S0 * \neg S1 * D0) + (\neg S0 * S1 * D1) + (S0 * \neg S1 * D2) + (S0 * S1 * D3)$

Row 1: $(\neg S0 + \neg S1 + \neg D0) = 0$

Row 3: $(\neg S0 + S1 + \neg D1) = 0$

Row 5: $(S0 + \neg S1 + \neg D2) = 0$

Row 7: $(S0 + S1 + \neg D3) = 0$

POS $\rightarrow Y = (\neg S0 + \neg S1 + \neg D0) * (\neg S0 + S1 + \neg D1) * (S0 + \neg S1 + \neg D2) * (S0 + S1 + \neg D3)$

Table 6: Voltage Table for 4-input MUX with wildcards.

S0(H)	S1(H)	D0(L)	D1(H)	D2(H)	D3(H)	Y(H)
L	L	L	-	-	-	H
L	L	H	-	-	-	L
L	H	-	L	-	-	L
L	H	-	H	-	-	H
H	L	-	-	L	-	L
H	L	-	-	H	-	H
H	H	-	-	-	L	L
H	H	-	-	-	H	H

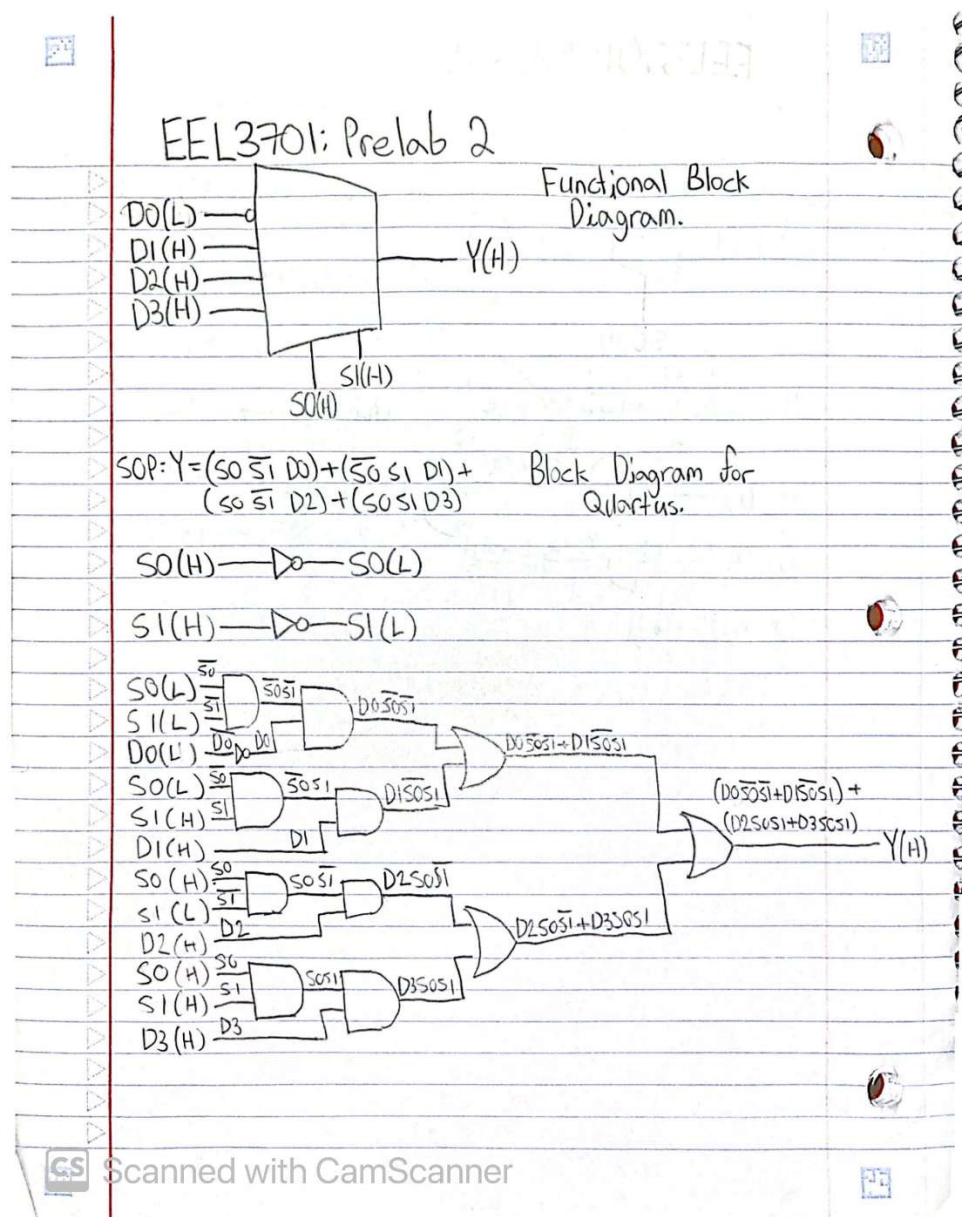


Figure 5: Functional Block Diagram and Circuit Design for 4-input MUX.

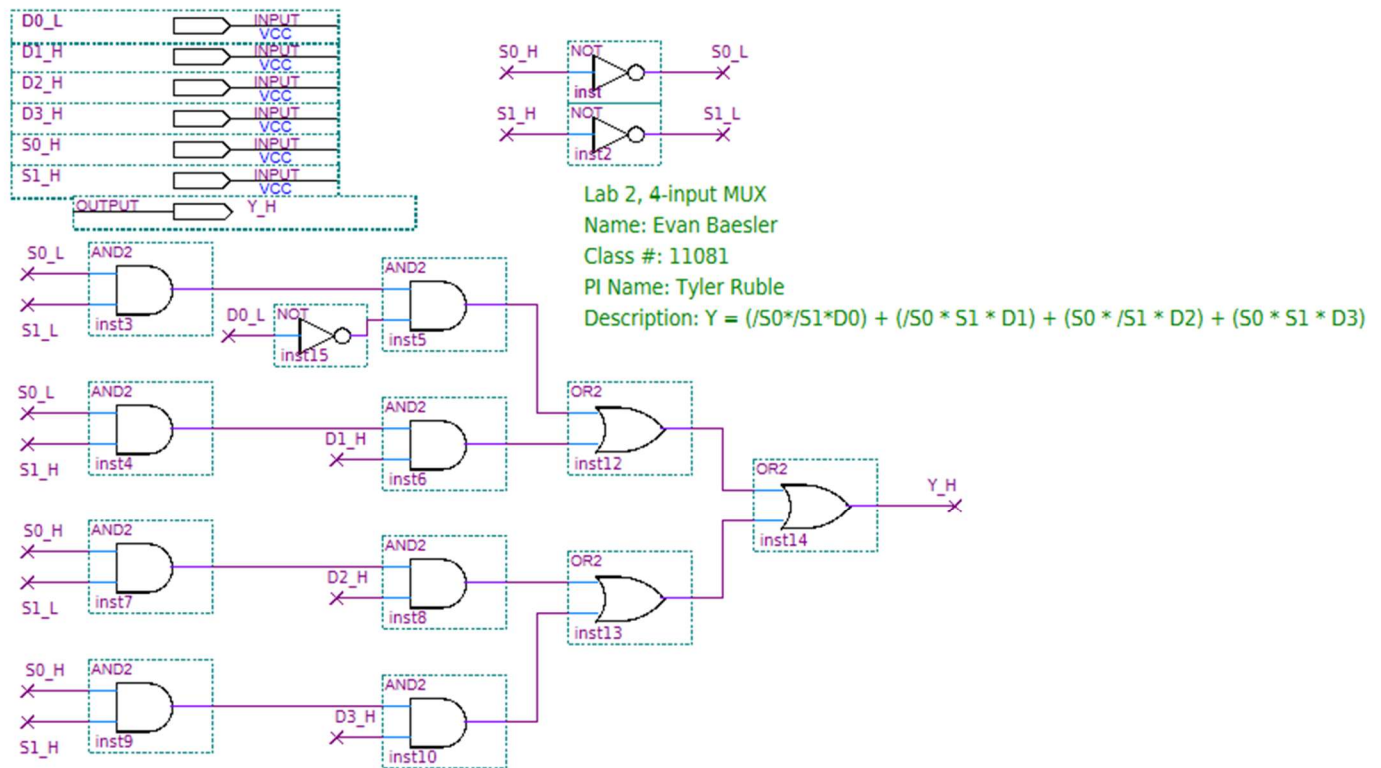
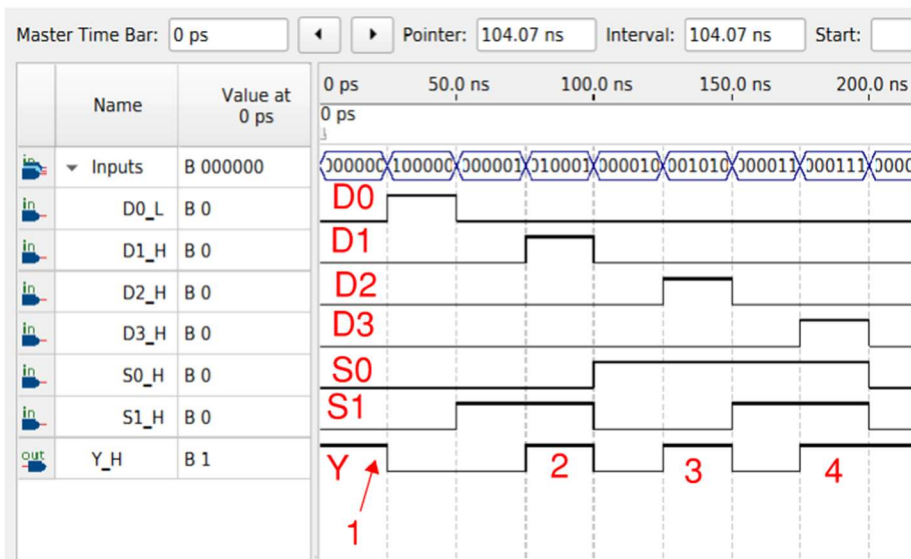


Figure 6: Quartus Circuit Diagram for a 4-input MUX.



Following our voltage table, we have a voltage high at values:

- 1) $/S0*/S1*/D0$ <-- (because D0(L))
- 2) $/S0*S1*D1$
- 3) $S0*/S1*D2$
- 4) $S0*S1*D3$

Figure 7: Quartus Waveform Simulation for 4-input MUX.

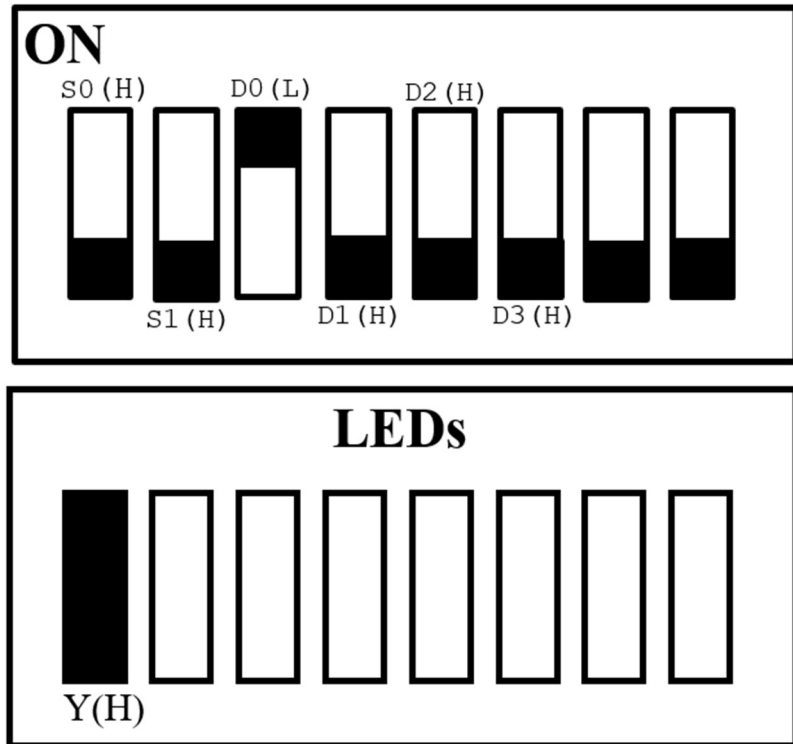


Figure 8: LED & Switch Legend for 4-input MUX.

SECTION 3

Table 7: Truth Table for Hex to 7-Segment Decoder

T	X3	X2	X1	X0	A	B	C	D	E	F	G	LEGEND
0	0	0	0	0	1	1	1	1	1	1	0	0
0	0	0	0	1	0	1	1	0	0	0	0	1
0	0	0	1	0	1	1	0	1	1	0	1	2
0	0	0	1	1	1	1	1	1	0	0	1	3
0	0	1	0	0	0	1	1	0	0	1	1	4
0	0	1	0	1	1	0	1	1	0	1	1	5
0	0	1	1	0	1	0	1	1	1	1	1	6
0	0	1	1	1	1	1	1	0	0	0	0	7
0	1	0	0	0	1	1	1	1	1	1	1	8
0	1	0	0	1	1	1	1	1	0	1	1	9
0	1	0	1	0	1	1	1	0	1	1	1	A
0	1	0	1	1	0	0	1	1	1	1	1	b
0	1	1	0	0	1	0	0	1	1	1	0	C
0	1	1	0	1	0	1	1	1	1	0	1	d
0	1	1	1	0	1	0	0	1	1	1	1	E
0	1	1	1	1	1	0	0	0	1	1	1	F
1	-	-	-	-	1	1	1	1	1	1	1	TEST

SOP(A):

$$T + (/T*/X0*/X1*/X2*/X3) + (/T*/X0*X1*/X2*/X3) + (/T*X0*X1*/X2*/X3) + (/T*X0*/X1*X2*/X3) + (/T*X0*X1*X2*/X3) + (/T*X0*X1*X2*/X3) + (/T*/X0*/X1*/X2*X3) + (/T*X0*/X1*/X2*X3) + (/T*/X0*X1*/X2*X3) + (/T*X0*X1*X2*X3) + (/T*/X0*X1*X2*X3) + (/T*/X0*/X1*X2*X3) + (/T*X0*X1*X2*X3) + (/T*X0*X1*X2*X3)$$

POS(A):

$$(T+/X0+X1+X2+X3)*(T+X0+X1+/X2+X3)*(T+/X0+/X1+X2+/X3)*(T+/X0+X1+/X2+/X3) \\ = T*/(X0+X1+X2+X3)*(X0+X1+/X2+X3)*(X0+/X1+X2+/X3)*(X0+X1+/X2+/X3)$$

SOP(B):

$$T + (/T*/X0*/X1*/X2*/X3) + (/T*X0*/X1*/X2*/X3) + (/T*/X0*X1*/X2*/X3) + (/T*X0*X1*/X2*/X3) + (/T*/X0*/X1*X2*/X3) + (/T*X0*X1*X2*/X3) + (/T*/X0*/X1*/X2*X3) + (/T*X0*X1*/X2*X3) + (/T*/X0*X1*X2*X3) + (/T*X0*X1*X2*X3)$$

POS(B):

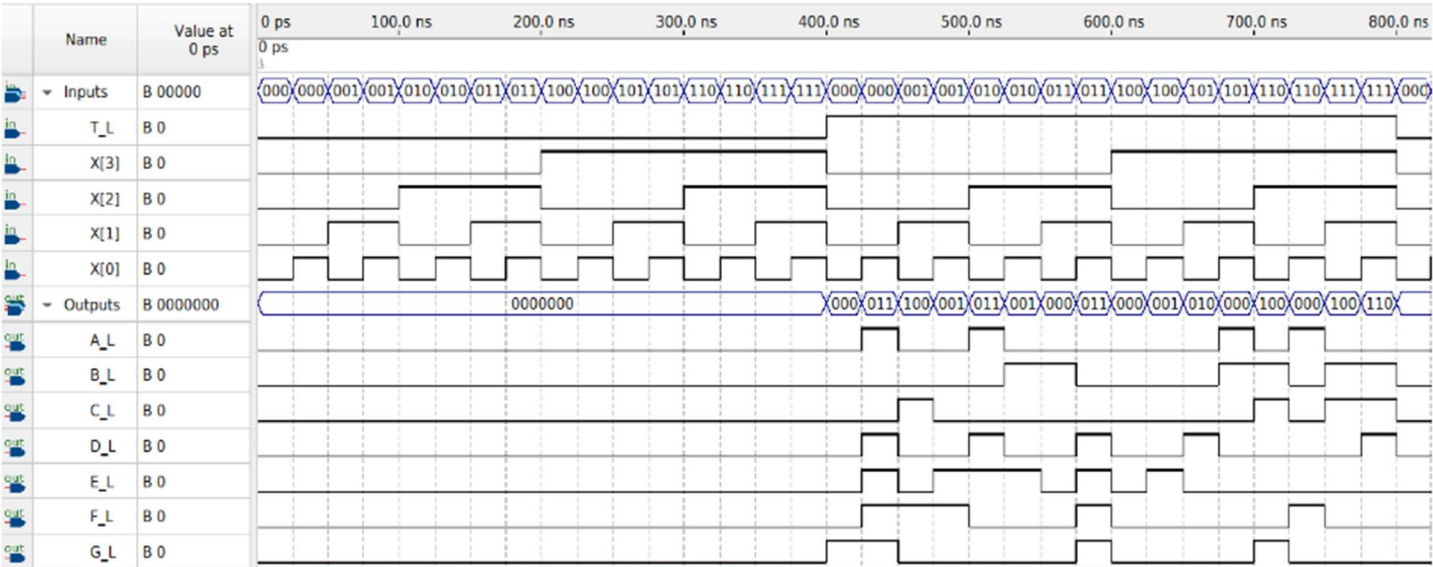
$$(T + /X0 + X1 + /X2 + X3) * (T + X0 + /X1 + /X2 + X3) * (T + /X0 + /X1 + X2 + /X3) * (T + X0 + X1 + /X2 + /X3) * (T + X0 + /X1 + /X2 + /X3) * (T + /X0 + /X1 + /X2 + /X3) \\ = T * (/X0 + X1 + /X2 + X3) * (X0 + /X1 + /X2 + X3) * (/X0 + /X1 + X2 + /X3) * (X0 + X1 + /X2 + /X3) * (X0 + /X1 + /X2 + /X3) * (/X0 + /X1 + /X2 + /X3)$$

SOP(C):

$$T + (/X0*/X1*/X2*/X3) + (X0*/X1*/X2*/X3) + (X0*X1*/X2*/X3) + (/X0*/X1*X2*/X3) + (X0*/X1*X2*/X3) + (/X0*X1*X2*/X3) + (X0*X1*X2*/X3) + (/X0*/X1*/X2*X3) + (X0*/X1*/X2*X3) + (/X0*X1*/X2*X3) + (X0*X1*/X2*X3) + (X0*/X1*X2*X3) + (X0*X1*X2*X3)$$

Table 8: Voltage Table for 4-input MUX

T(L)	X3(H)	X2(H)	X1(H)	X0(H)	A(L)	B(L)	LEGEND
L	-	-	-	-	L	L	TEST
H	L	L	L	L	L	L	0
H	L	L	L	H	H	L	1
H	L	L	H	L	L	L	2
H	L	L	H	H	L	L	3
H	L	H	L	L	H	L	4
H	L	H	L	H	L	H	5
H	L	H	H	L	L	H	6
H	L	H	H	H	L	L	7
H	H	L	L	L	L	L	8
H	H	L	L	H	L	L	9
H	H	L	H	L	L	L	A
H	H	L	H	H	H	H	b
H	H	H	L	L	L	H	C
H	H	H	L	H	H	L	d
H	H	H	H	L	L	H	E
H	H	H	H	H	L	H	F



TOGGLE SECTION 0123456789AbCdEF0

Since A and B are both active-low, we can prove our voltage table by acknowledging that A should only be off (H) for 1, 4, b, and d. We can do the same for b by acknowledging B should only be off (high) for 5, 6, b, C, E, F.

Since the lights are all active-low, we can also acknowledge that the toggle section should be set to on (L) to turn on.

Figure 9: Annotated 7-Segment Display Quartus Waveform Simulation.

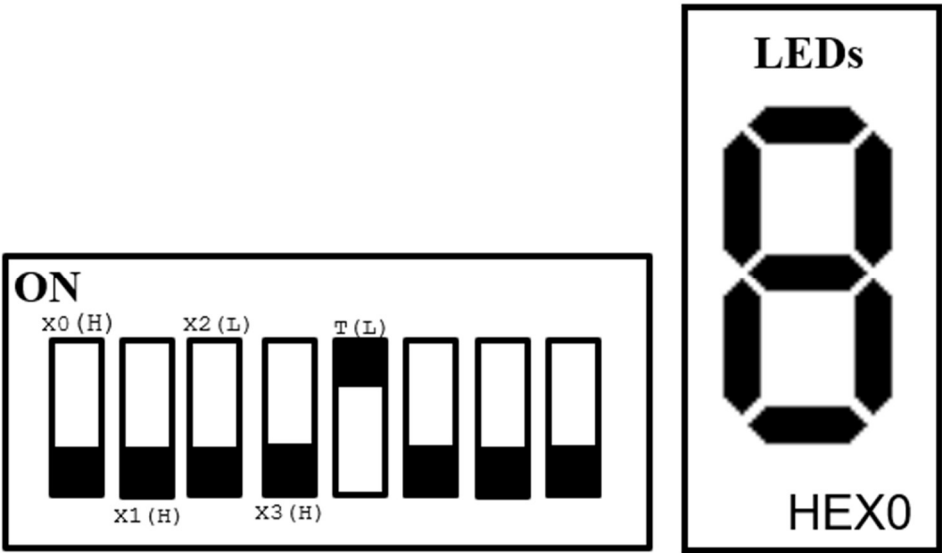


Figure 10: Switch & LED Legends for Hex to 7-Segment Display

SECTION 4

Lab 2, MUX Testing
Name: Evan Baesler
Class #: 11081
PI Name: Tyler Ruble
Description: 74'153 MUX Testing to see if A & B = S0 & S1 or S1 & S0

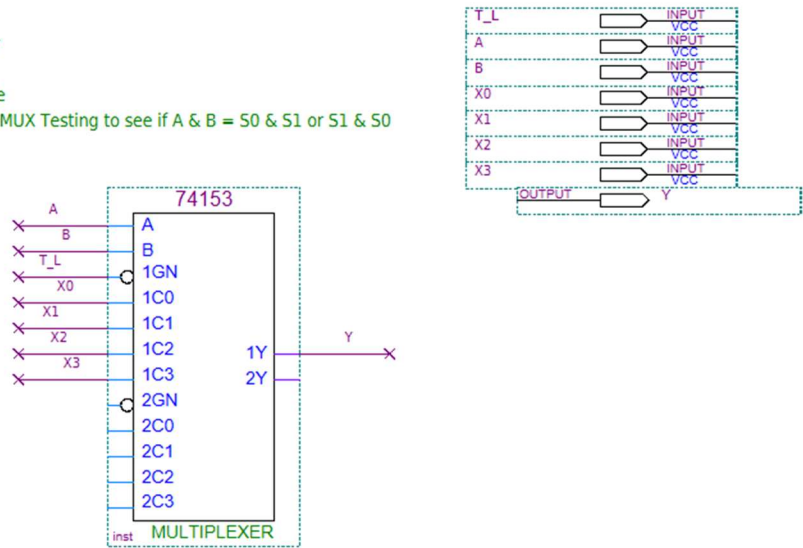
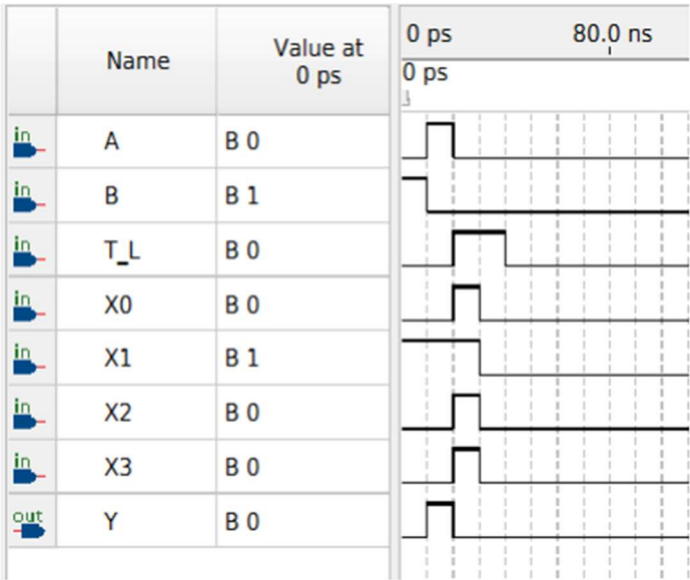


Figure 11: Quartus Diagram of 74153 Testing



We can see that A = 1 connects X1 to Y, meaning: A = S0, B = S1.

We also see that when T_L is high that our Y is low.

Figure 12: Test of 74'153 in Quartus Waveform.

Formula: $F = W(\overline{X} + XYZ) + \overline{W}(\overline{W} + Z)$

Table 9: Truth Table for $F = W(\overline{X} + XYZ) + \overline{W}(\overline{W} + Z)$

W	X	Y	Z	$\overline{X}$	XYZ	$\overline{X} + XYZ$	$W(\overline{X} + XYZ)$	$\overline{W}$	$\overline{W} + Z$	$\overline{W}(\overline{W} + Z)$	F
0	0	0	0	1	0	1	0	1	1	0	0
0	0	0	1	1	0	1	0	1	1	0	0
0	0	1	0	1	0	1	0	1	1	0	0
0	0	1	1	1	0	1	0	1	1	0	0
0	1	0	0	0	0	0	0	1	1	0	0
0	1	0	1	0	0	0	0	1	1	0	0
0	1	1	0	0	0	0	0	1	1	0	0
0	1	1	1	0	1	1	0	1	1	0	0
1	0	0	0	1	0	1	1	0	0	1	1
1	0	0	1	1	0	1	1	0	1	0	1
1	0	1	0	1	0	1	1	0	0	1	1
1	0	1	1	1	0	1	1	0	1	0	1
1	1	0	0	0	0	0	0	0	0	1	1
1	1	0	1	0	0	0	0	0	1	0	0
1	1	1	0	0	0	0	0	0	0	1	1
1	1	1	1	0	1	1	1	0	1	0	1

SOP:

$$(W \cdot \overline{X} \cdot \overline{Y} \cdot \overline{Z}) + (W \cdot \overline{X} \cdot \overline{Y} \cdot Z) + (W \cdot \overline{X} \cdot Y \cdot \overline{Z}) + (W \cdot \overline{X} \cdot Y \cdot Z) + (W \cdot X \cdot \overline{Y} \cdot \overline{Z}) + (W \cdot X \cdot Y \cdot \overline{Z}) + (W \cdot X \cdot Y \cdot Z)$$

POS:

$$(W + X + Y + Z) \cdot (W + X + Y + \overline{Z}) \cdot (W + X + \overline{Y} + Z) \cdot (W + X + \overline{Y} + \overline{Z}) \cdot (W + \overline{X} + Y + Z) \cdot (W + \overline{X} + Y + \overline{Z}) \cdot (W + \overline{X} + \overline{Y} + Z) \cdot (W + \overline{X} + \overline{Y} + \overline{Z})$$

SOP K-MAP

Table 10: K-Map to simplify function F

WX \ YZ	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	0	1	1
10	1	1	1	1

$W \cdot \overline{Z}$

$W \cdot \overline{X}$

$W \cdot Y$

$$F = W(\overline{X} + Y + \overline{Z})$$

Table 11: Truth Table for Simplified $F = W(X+Y+Z)$

W	X	Y	Z	/X	/Z	/X+Y+/Z	F
0	0	0	0	1	1	1	0
0	0	0	1	1	0	1	0
0	0	1	0	1	1	1	0
0	0	1	1	1	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	0	0	0
0	1	1	0	0	1	1	0
0	1	1	1	0	0	1	0
1	0	0	0	1	1	1	1
1	0	0	1	1	0	1	1
1	0	1	0	1	1	1	1
1	0	1	1	1	0	1	1
1	1	0	0	0	1	1	1
1	1	0	1	0	0	0	0
1	1	1	0	0	1	1	1
1	1	1	1	0	0	1	1

Table 12: Voltage Table for Simplified $F = W(X+Y+Z)$

W(L)	X(H)	Y(H)	Z(H)	F(H)
L	L	L	L	H
L	L	L	H	H
L	L	H	L	H
L	L	H	H	H
L	H	L	L	H
L	H	L	H	L
L	H	H	L	H
L	H	H	H	H
H	L	L	L	L
H	L	L	H	L
H	L	H	L	L
H	L	H	H	L
H	H	L	L	L
H	H	L	H	L
H	H	H	L	L
H	H	H	H	L

Lab 2, MUX Testing
Name: Evan Baesler
Class #: 11081
PI Name: Tyler Ruble
Description: 74'153 MUX to integrate $F = W(X + XYZ) + /(W + Z)$

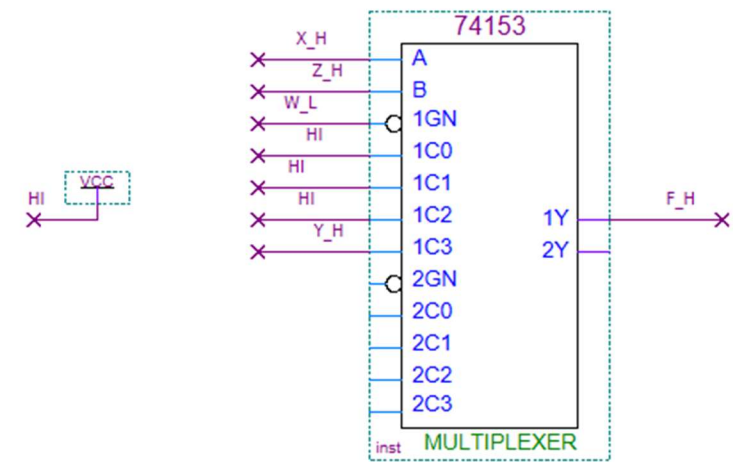
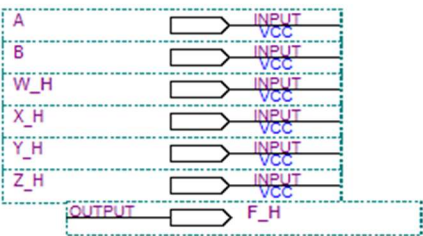
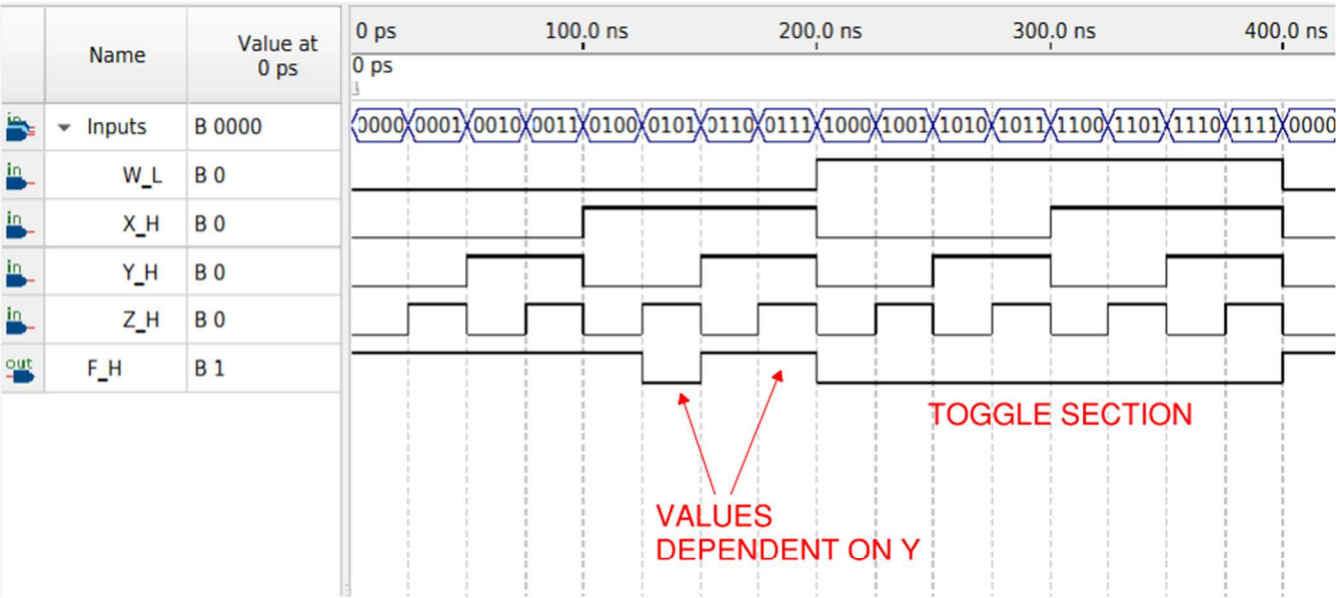


Figure 13: Quartus Diagram for $F = W/(X+Y+Z)$



We can see from our voltage table that we are matching the one we calculated. We can also see that the other values are high for when either X or Z are low (X0X0, X0X1, X1X0). We also see that when W is activated our toggle is engaged.

Figure 14: Annotated Quartus Waveform for 4-input MUX and $F = W/(X+Y+Z)$